

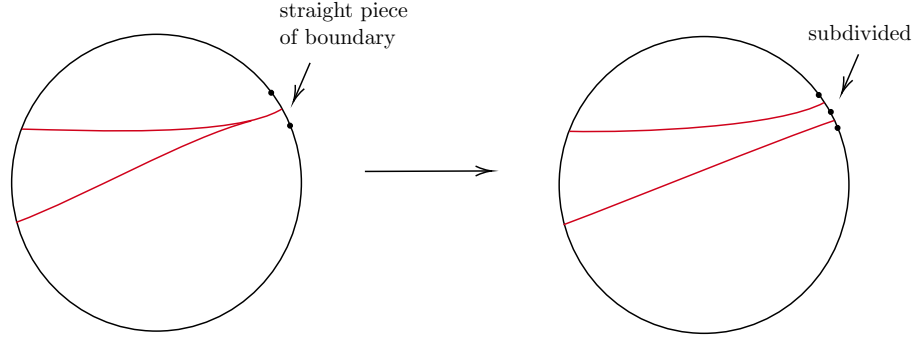


Figure 29: When the slide reaches the boundary.

3 Leaf properties

Having established the disk diagram theorems of the previous section, we will now use them to prove various properties of leaves. We assume here that M is a simply connected completely locally cubulated. In light of Theorem 2.1, we may assume that all disk diagrams for loops are standard and minimal.

Germ. We will use the term *edge* to refer to a 1-cube, not necessarily a local 1-cube. Suppose that $a \in M$ is a point and e is an edge with $a \in e$. We call this a *pointed edge* (e, a) . Two such pointed edges (e, a) and (f, a) are *germ equivalent* if $e \cap f$ is a non-degenerate sub-edge of both e and f (i.e. not a point) and a is in the interior of $e \cap f$ or a is a boundary point of both e or f . We call such an equivalence class a *germ* of an edge at a . We will denote germ by (ϵ, a) . Two such germs (ϵ_1, a) and (ϵ_2, b) are said to be *elementarily equivalent* if there exists a local square σ such that some representative of ϵ_1 of ϵ_1 and ϵ_2 of ϵ_2 are opposite edges of σ and a and b are on the same plaque of σ . The equivalence relation generated by elementary equivalence is called *parallelism*. Thus two germs are parallel if there exists a finite sequence of elementary equivalences between them. We denote by $[\epsilon, a]$ the parallelism class of (ϵ, a) . If (f, b) is a pointed edge whose germ is parallel to (ϵ, a) , then will also write $[f, b]$ for the equivalence class $[\epsilon, a]$.

Leaves. Given a germ (ϵ, a) , we let $L_{\epsilon, a} = \{b \mid [f, b] = [\epsilon, a]\}$. Notice that given local cube σ , and edge e of σ and a point $a \in e$ such that $(e, a) \in [\epsilon, a]$, the unique plaque containing a is contained in $L_{\epsilon, a}$. This is not hard to see because subcubes of local cubes are local cubes.

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We call $L_{\epsilon,a}$ the *leaf* dual to the germ class (ϵ,a) . We call the leaf *bicollared* when the germ (ϵ,a) has the property that $b \in \text{Int}(f)$ for any $(f,b) \in [\epsilon,a]$ representing the germ. Otherwise, we say the leaf is *collared*, which is to say that $f \in \partial b$ for any $(f,b) \in [\epsilon,a]$.

Finally a bicollared leaf $L_{\epsilon,a}$ will be said to be *regular* if every point on $L_{\epsilon,a}$ has a cubulated neighborhood which is of the form $N \cong K \times [-1,1]$, where K is a CAT(0) cube complex with a single vertex and $L \cong K \times \{0\}$. Thus, when $L_{\epsilon,a}$ is regular, it locally separates.

We wish to establish several properties of leaves. Particularly regular leaves and collared leaves will play a special role.

We start with a local lemma.

Lemma 3.1. *Let L be a leaf. If σ is an n -cube containing n adjacent edges all of whom are in L , then $\sigma \subset L$*

Proof. We first prove the claim for $n = 2$. We let f_1 and f_2 be those adjacent edges and $a = f_1 \cap f_2$. Since $a \in L$, there exists a local edge e such that the pointed edge (e,a) is in the parallelism class to which L is dual, so that $L = L_{e,a}$. We consider a point $p \in \sigma$. We wish to show that $p \in L$, so we may assume $p \notin f_1 \cup f_2$. Then p and a span a subsquare of σ . So it is enough to show the claim when the point p is the diagonally opposite point to a in σ .

We will show the claim when L is collared. The case when L is bicollared is similar. For $i = 1, 2$, let a_i be the endpoint of e_i that is not a . Then since $f_i \in L$, there exists a pointed edge (e_i, b) and a sequence of squares $\tau_1^i, \dots, \tau_n^i$ realizing the parallelism between $[e, a] = [e_i, a_i]$. Note that this may require passing to subedges of the pointed edges (e, a) and (e_i, a_i) in the same respective germ classes. After subdividing, we may assume that the τ_j^i 's match up so that the union of the τ_j^i 's give two bands of squares, B_i joining (e, a) and (e_i, a_i) , for $i = 1, 2$. See Figure 30. The subdivision of B_i into squares, gives a subdivision of e_i into edges and taking the product yields a subdivision of σ into subsquares, σ_{ij} where σ_{11} is a square containing a . Now note that $\tau_1^1, \tau_1^2, \sigma_{11}$ are a cycle of three squares at a . By the 3-cube assumption, there exists a 3-cube of which these 3 squares are faces. Having glued in this 3-cube, we will see new points and cycles of squares involving other τ_j^i 's, σ_{ij} 's and faces of cubes already glued. We continue to fill in these 3-cubes until there are no more missing 3-cubes. This will now produce a sequence of squares joining (e_1, a_1) and an edge (e', p) . See Figure 30

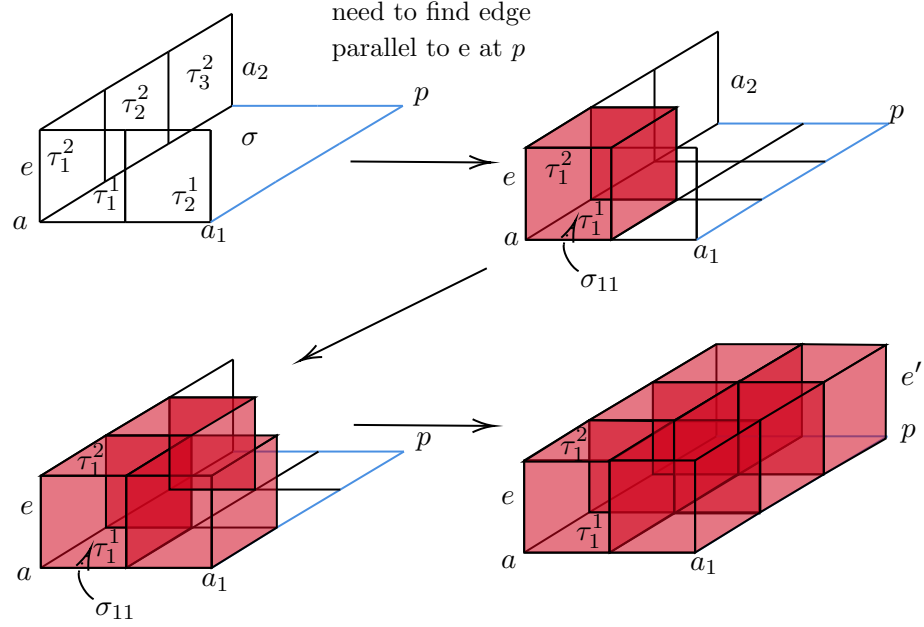


Figure 30: Filling in the cubes to get an a parallel edge to e at p .

For $n > 2$, we proceed by induction. For the inductive step, let a be a vertex of σ contained the edges of $e_1, \dots, e_n$. We will show that the diagonally opposite vertex to a in σ is in L . This will suffice to show the claim, since every point of σ is the diagonally opposite point of a in some subcube of σ containing a and spanned by subedges of $e_1, \dots, e_n$.

We coordinatize σ so that $a = (0, \dots, 0)$. We are given that all the vertices of the form $a_i = (0, \dots, 0, \overset{i}{1}, 0, \dots)$ are in L as well. By induction, we then have that all the $(n-1)$ -cubes at a are in L . This tells us that the edges $[(1, \dots, 1, 0, 0), (1, \dots, 1, 1, 0)]$ and $[(1, \dots, 1, 0, 0), (1, \dots, 1, 0, 1)]$ are in σ . These are adjacent edges in a square containing the point $(1, \dots, 1)$, so by the claim for $n = 2$, we get that $(1, \dots, 1) \in L$, as required. $\square$

Lemma 3.2. *Let $L_{\epsilon, a}$ be a leaf and e be a 1-cube (not necessarily local). Then e contains at most one point b such that $[e, b] = [\epsilon, a]$*

Proof. Suppose that there exists two such points $b, c \in e$. By definition, we thus have a sequence of local squares $\sigma_1, \dots, \sigma_n$ which carry out the parallel equivalence. $\square$

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Old stuff. Now returning to our locally cubulated space M , a *leaf path* α joining two points x and y is a path in M with the following properties. There exists a finite collection of cubulated neighborhoods $N_1, \dots, N_m$ and a partition of α into subpaths $\alpha = \alpha_1 \cdot \alpha_2 \cdot \dots \cdot \alpha_m$, such that $\alpha_i \in \text{int}(N_i)$ and α_i is contained in a regular leaf of N_i . Given a plaque π in M , the *leaf* containing π is the subset of points of M that can be joined to a point in π by a leaf path whose first segment lies in π . In particular, a point in the interior of a local edge determines a leaf, since it is a plaque of the edge.

We will now define regular leaves. First, we define regular leaves for a CAT(0) cube complex X . For a hyperplane H of X , let $N(H)$ denote the union of the cubes intersecting H non-trivially. Recall that $N(H)$ has a product structure of H with a unit interval. A *regular leaf* is a parallel copy of H in $\text{int}(N(H))$ for some hyperplane of H . More precisely, in the identification $N(H) \cong H \times [0, 1]$, a regular leaf is a subset of the form $H \times \{t\}$ for $0 < t < 1$.

Note also that if N is some cubulated neighborhood and π is a regular plaque of some cube of N , then the entire regular leaf of N containing that plaque is contained in the leaf of M determined by that plaque.

Just as for plaques, unless stated otherwise, we will use the term “leaf” for a codimension-1 leaf. A consequence of the definition is that if L is a leaf that intersects a cube of M in a plaque that is transverse to some edge e at some point p , then for all cubes that contain an edge with p in its interior, the leaf extends into those cubes as well.

old definition: A *regular leaf* is a leaf L that does not contain a singular point of M .

A leaf is *regular* if it locally has a transverse interval with which it spans a product. More explicitly, let Y be a finite union of cubes. Then a leaf is regular in Y if its intersection with each cube of Y is a reg

This implies first of all that at every point $x \in L$, there exists a 1-cube e with $x \in \text{int}(e)$ such that L meets e transversely at x . But it also means that there exists a neighborhood $N \cong N' \times e$ such that in this identification with a product $x \in N' \times \{t\} \subset L$. Since the collection of singular points is countable, regular leaves are generic. We will focus for now on regular leaves and define singular leaves later.

The first thing we will see is that regular leaves do not self-intersect. In fact, we have the following.

Lemma 3.3. *Let L be a regular leaf and σ a local cube. Then $\sigma \cap L$ is a single plaque.*

Proof. Suppose that π_1, π_2 are two plaques in $\sigma \cap L$. There are two cases, depending on whether π_1 and π_2 intersect or not. Let us first consider the case that π_1 and π_2 intersect. In this instance, there exists a 2-face τ of σ such that $\pi_1 \cap \tau$ and $\pi_2 \cap \tau$ are two intersecting 1-dimensional plaques in L . By definition, we have a leaf path joining the plaques $\pi_1 \cap \tau$ and $\pi_2 \cap \tau$. This means that, after possibly passing to a subsquare of τ , we have a band of squares joining two adjacent edges e and f of τ . We then have the two possibilities pictured in Figure 31.

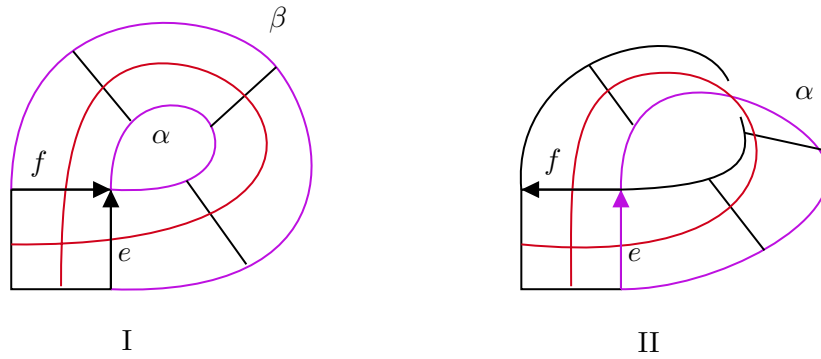
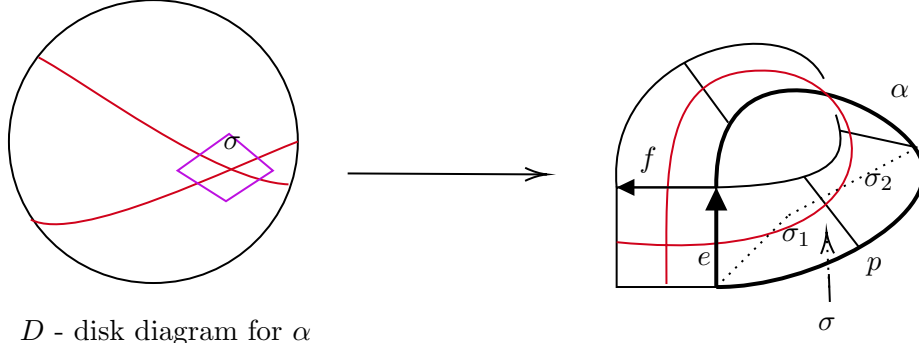


Figure 31: Two ways a regular leaf can self-intersect.

For the type I intersection, we consider a disk diagram D for the loop α . We then attach to this the band of squares between α and β to produce a disk diagram D' for β . This disk diagram D' has a 1-loop in it. Thus, as in the proof of the disk diagram theorem, after finitely many bigon and triangle moves, we can produce a standard disk diagram with a 1-gon, which is impossible.

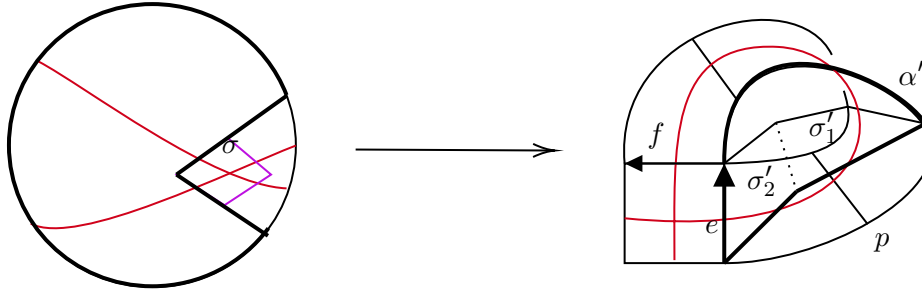
For the type II intersection, we consider a minimal disk diagram for the loop α indicated in Figure 31. Amongst all such configurations (square of self intersection, band with half twist, loop α subdivided so that it has a standard minimal disk diagram), we choose one such that the complexity $(length(\alpha), area(D))$ is minimal. We note that by the minimality of $length(\alpha)$, there is no backtracking along α , so that the disk D has no spikes. It follows that D has three corners. We can thus assume that there is a corner that is along some point p along α that is not an endpoint of e . Let σ_1 and σ_2 be the two squares of the band that have p as a vertex. The corner of D at p provides a square σ at p that is the image of a square σ in the vK-diagram of D . (See Figure 32.)



D - disk diagram for α

Figure 32: The corner of the disk diagram provides a new square at p .

By completeness of the local cubulation structure, there exists a 3-cube τ at p which σ_1, σ_2 and σ as faces. Let σ'_1 and σ'_2 be the faces opposite of σ_1 and σ_2 respectively. Replacing σ_1 and σ_2 with σ'_1 and σ'_2 gives us a new band and a new path α' . We also obtain a new disk D' obtained from D by cutting out the square σ in D . We thus obtain a new configuration with a disk diagram of smaller area, and hence complexity, a contradiction. (See Figure 33.)



D' - a new disk diagram for α'

Figure 33: Replacing the band with a new band and a smaller disk diagram.

We now need to address the case that the two plaques π_1 and π_2 of σ do not self intersect. This is done in a similar way as the previous case of intersecting plaques. As before, we have a band of squares joining two sub-edges of edges e and f of σ , except that this time e and f are parallel. Note that since e and f are parallel, the band can be extended inside σ to join two subedges of the same edge e . Thus, we can reduce our discussion

to the 1-dimensional cube e . As before we have bands joining the endpoints of the plaques, which either give an annulus or a mobius band. See Figure 34.

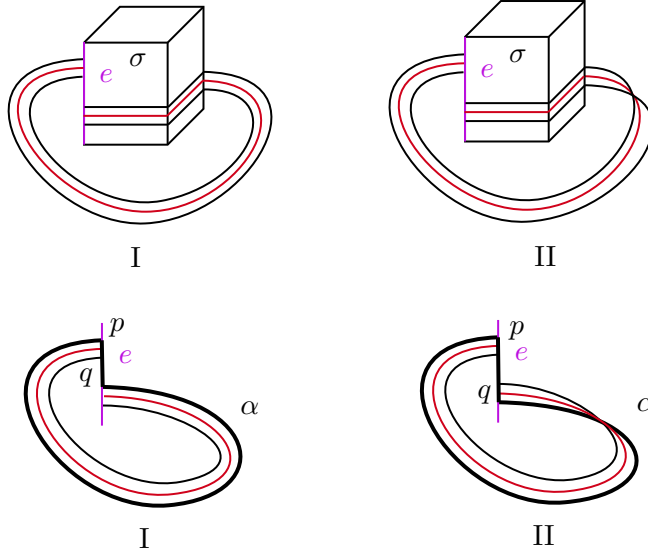


Figure 34: The case of non-intersecting plaques.

In both cases, we can take a loop α as before and reduce the area of a disk bounding it by finding a corner along α that is not at the points p or q , as shown. This corner cannot be along e since e is straight, so it must be along the band. As the argument in the crossing case, a corner move produces a configuration of the same type of lower complexity, a contradiction. $\square$

Much as hyperplanes in a CAT(0) cube complex have a natural CAT(0) cubical structure inherited from the ambient space, a regular leaf L inherits a local cubulation from M . Explicitly, consider a regular leaf L and a point $x \in L$. As noted above, since L is regular, there exists an edge e transverse to L at x . By definition x is in the interior of a cubulated neighborhood N such that the intersection of N with L contains the hyperplane

, there is a neighborhood N has a (different) product cubical structure $N \cong N_0 \times [0, 1]$, where N_0 is also a CAT(0) cube complex of with a central vertex x_0 , so that “horizontal” plaques of the form $\sigma \times \{t_0\}$ are plaques of the leaf L , for some $0 < t_0 < 1$. The intersection of L with each cube of $N \cong N_0 \times [0, 1]$ then locally endows L with a cube complex structure

isomorphic to the hyperplane of N transverse to $\{x_0\} \times [0, 1]$. As hyperplanes of CAT(0) cube complexes are CAT(0), we get that this local cubulation is CAT(0).

Together with the regularity condition, we will now see that in compact regions, a regular leaf behaves much like a hyperplane in that it has a trivially foliated regular neighborhood.

To this end, it is useful to describe a transverse holonomy on leaves. Let α be a piecewise straight path in a regular leaf L joining points a and b . We want to describe a map between an arc transverse to L at a and one transverse to L at b . To start with, if a and b are in the same plaque $\pi \subset L$ of an n -cube σ , then σ has a product structure of an $n - 1$ cube cross a 1-cube $\sigma = H_\sigma \times I_\sigma$, where a and b have the same I_σ coordinate. (We view σ_H as the horizontal factor and I_σ as the vertical factor.) If $p : \sigma \rightarrow H_\sigma$ is the projection map, then $I_a = p^{-1}(p(a))$ is called the vertical transversal through a associated with the cube σ . We then have a natural bijection $\eta : I_a \rightarrow I_b$ which identifies points in the same plaques parallel to P . Note that if we kept the same points a and b and changed the ambient cube σ , we might get different transversals but any two such intersect in a transversal, so what is well defined is a germ of a transverse arc to a regular leaf L at a point.

Suppose now that L is a leaf which runs through the interior of two intersecting cubes σ and τ . Suppose that a and b are two points in L , one in σ and one in τ . Then the transverse arc at a in σ and the transverse arc at b in τ might not have an identification between them as above. However, we can find a subtransversal and a natural identification. More precisely, suppose that $\sigma = H_\sigma \times I_\sigma$ and $\tau = H_\tau \times I_\tau$. Let $c \in L \cap \sigma \cap \tau$. Then we obtain natural identifications of the transversals in $H_\sigma \times (I_\sigma \cap I_\tau)$ to a and c in the cube $H_\sigma \times (I_\sigma \cap I_\tau)$. Similarly we get natural identifications between the transversal to b and to c in the cube $H_\tau \times (I_\sigma \cap I_\tau)$. The composition gives an identification between germs of transversals at a and b which identify points in the same leaves. By induction, given a leaf path α joining two points in the $a, b \in L$, we obtain a natural identification between transversals at a and b , $\eta_\alpha : I_a \rightarrow I_b$.

Now given a loop α we obtain a map $\eta_\alpha : I_a \rightarrow I'_a$ between two transversals at a . We call this the *holonomy map*.

Lemma 3.4. *The holonomy map under any leaf path α is the identity.*

Proof. Suppose that α is a leaf path and $x \in I_a$, such that $\eta_\alpha(x) \neq x$. Note that η_α identifies points in the same leaf. Thus we obtain x and $\eta_\alpha(x)$ lie

in the same leaf. But $\eta_\alpha(x) \neq x$ means that x and $\eta_\alpha(x)$ are in different plaques in the same cube, a contradiction to Lemma ??.

□

We are now ready to describe regular neighborhoods of regular leaves.

Lemma 3.5. *Let L be a regular leaf in M . Let $K \subset L$ be a finite union of cubes of M . Let $L_K = L \cap K$. Then there exists an embedding $\iota : L_K \times I \rightarrow K$ such that $\iota(L' \times \{x\})$ is a regular leaf in K .*

Proof. Let $K = \cup_{i=1}^n C_i$ a finite union of cubes. We may assume that L_K and K are connected, for otherwise we consider each component separately. We prove the lemma by induction on n . For $n = 1$ the claim is clear. Suppose that the lemma is true for $n - 1$. Let $K' = \cup_{i=1}^{n-1} C_i$. Then we have an embedding $\iota : L_{K'} \times I \rightarrow K'$ sending horizontal regular leaves to regular leaves in K' . Let $N = \text{Im}(\iota)$. We now consider the intersection of N with C_n . Let $P = L \cap C_n$, which by Lemma ?? is a single plaque. Now each component D_i of $N \cap C_n$ is a subcube of C_n of the form $D_j = H_j \times I_j$. By Lemma 3.4 $I_j = I_k$ for any two such components. It follows that after possibly passing to a further subinterval of I , we can extend ι from $L_{K'} \times I$ to $L_K \times I$. □

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We now establish further properties of regular leaves.

Proposition 3.6.

1. *Each regular leaf separates M into two complementary components.*
2. *Each regular leaf is simply connected.*
3. *The intersection of two regular leaves is either empty or a regular codimension-2 leaf.*
4. *For $n \geq 3$. Let $L_1, \dots, L_n$ be a collection of regular leaves such that $L_i \cap L_j \neq \emptyset$ for every i, j . Then $\cap_i L_i \neq \emptyset$.*

Proof.

1. To show the separation property, let L be a regular leaf and let e be a 1-plaque transverse to a plaque $P \subset L$. Let p and q denote the endpoints of e , we first show that p and q are in separate components. Suppose that there exists a path γ joining p and q in the complement of L . We homotop γ so that it is piecewise straight. The path γ together with e gives us a piecewise straight loop $\alpha = e \cup \gamma$ which crosses L only in an interior point x of e . We consider a disk diagram $f : D \rightarrow M$ for α . Since D is compact, $f(D)$ is compact. Let K be a finite union of cubes that contains $f(D)$. By

Lemma 3.5, $L_K = L \cap K$ has a regular neighborhood $N(L_K)$ consisting of leaves parallel to L_K . Since γ does not meet L_K , we can choose $N(L_K)$ so that γ does not intersect $N(L_K)$ as well.

Now we consider the point x . Recall that the process of producing a disk diagram for D involves possibly subdividing α . Thus x may be a vertex of α or a point in the interior of an edge of α . In the latter case, the leaf in D associated to x lies in $f^{-1}(L)$. Since this leaf is an arc with both endpoints on α , it follows that there is a point in γ mapped to L , a contradiction. If x is a vertex of α , we note that $f^{-1}(\text{int}(N(L_K)))$ is open and hence contains points in e close to x , but in the interior of an edge. Such a point is in the preimage of a leaf L' in $N(L_K)$. Again, now there is an arc of D in $f^{-1}(L')$. However, since by Lemma 3.5 $L_K = L \cap K$ has a regular neighborhood of leaves parallel to L_K , we have a leaf in D , that lies in the preimage of one such leaf L' . But since the leaves of D are arcs with endpoints on the boundary, there must be another edge in ∂D that crosses the preimage of L' . It follows that the path α crosses L at a point of γ , a contradiction.

Having shown that p and q are in different components, we now show that there are exactly two components. Given any point in $y \in M - L$, we join it by a piecewise straight path γ to L . We can choose y to be the first point of L that γ meets. So that γ meets L only at its endpoints. Now choose a piecewise straight path β joining y and the point $x \in e$. By Lemma 3.5, we have a regular neighborhood for some compact L_K containing β . We can then push β off of L . The endpoint of β will then be on the edge e and is on the same side as one of p or q . Thus there is a path in the complement of L from y to one of p or q .

2. To prove simple connectivity, let L be a regular leaf. Consider a loop α in L . We can homotop α in L so that it is piecewise straight. Now consider a minimal disk diagram D for α . We argue in a similar fashion to the proof of Lemma ???. Amongst all such α and D , with α non-trivial in $\pi_1(L)$, let $(\text{length}(\alpha), \text{area}(D))$ be minimal (in the lexicographical order). If α has backtracking, then we can reduce the length of α , so we may assume α has no backtracking. This means that D has three corners. In particular, there are two consecutive edges e and f along α that span a square σ . Since e and f are in L , so is σ . We then replace α with a path that contains the two other edges of σ . This is a path in L homotopic to α with a disk diagram of smaller area.

3. Next we turn to understanding the intersection of two regular leaves. Let L_1 and L_2 be regular leaves in M such that $L = L_1 \cap L_2 \neq \emptyset$. Note that the intersection of the two leaves is locally a codimension-2 leaf, so one just needs to prove that L is connected. Suppose that p and q are points lying

in separate components of $L_1 \cap L_2$. For $i = 1, 2$, let $\alpha_i \in L_i$ be a piecewise straight path joining p and q . We then obtain a loop $\alpha = \alpha_1 \cdot \bar{\alpha}_2$ be the loop that is the concatenation of α_1 and the inverse of α_2 . And let D be a minimal disk diagram for α . Let p' and q' be the points in D mapping to p and q respectively. Now by Lemma 2.2, there exists a corner or a spike along α_1 or α_2 . A spike will allow us to shorten the arc α_1 or α_2 and a corner allows us to reduce the area of D . We continue in this manner until we such a configuration with a disk D of area 0. Now if D had spikes along α_1 or α_2 , we reduce the length. Thus we obtain such a configuration in which D consists of embedded arcs separating the points p' and q' . It follows that $\alpha_1 = \alpha_2$, which yields a path in $L_1 \cap L_2$ joining p and q .

4. Finally, we turn to the last claim. We first prove it for $n = 3$. We consider a triple of regular leaves L_1, L_2, L_3 that pairwise intersect. Assume that $L_1 \cap L_2 \cap L_3 = \emptyset$. As in the proof of Lemma ??, we consider points $p_{ij} \in L_i \cap L_j$ for $i \neq j \in \{1, 2, 3\}$. As the leaves are regular, for each i, j there exist squares σ_{ij} containing p_{ij} in their interior. As in previous arguments, we want to connect these by bands and then build an appropriate disk diagram. We need to first show that it is a standard configuration as shown in Figure 35, and in particular the bands do not fit together to make a mobius band. To see this. As we have shown, $L_2 \cap L_3$ is connected. Thus, since it is disjoint from L_1 , there exists a complementary component L_1^+ of L containing $L_2 \cap L_3$. Similarly, for L_2 and L_3 , there exist complementary components L_2^+ and L_3^+ containing $L_1 \cap L_3$ and $L_1 \cap L_2$ respectively. Since $p_{ij} \in L_i \cap L_j \in L_k^+$ (for distinct i, j, k) and since our squares σ_{ij} can be chosen as small as we like, we can choose σ_{ij} such that so that $\sigma_{ij} \subset L_k^+$.

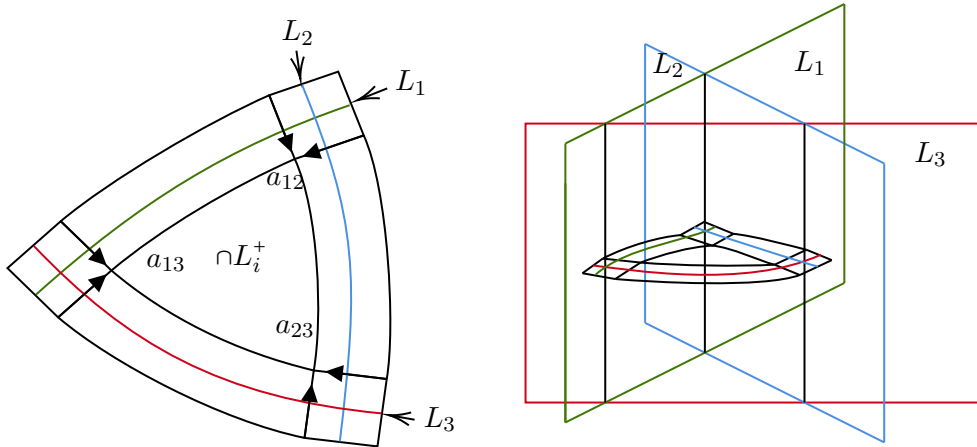


Figure 35: A triple of leaves that pairwise intersect.

Now note that each σ_{ij} has a unique vertex $a_{ij} \in L_i^+ \cap L_j^+$. It follows that $a_{ij} \in \cap_{i=1}^3 L_i^+$. For each edge of e that contains one of the a_{ij} 's we orient the edge towards the region $\cap_{i=1}^3 L_i^+$. Now we join the edges of σ_{ij} that contain the a_{ij} 's by foliated bands. The union of these bands and the σ_{ij} 's is then an annulus with one component of its boundary, α , containing the a_{ij} 's.

Let β and the other boundary component of the annulus. Consider a minimal disk diagram D' for α . Then D' together with the annulus, is a disk diagram for β . This disk diagram has three arcs $\alpha_1, \alpha_2, \alpha_3$ that form a 3-loop in D . We then can carry out bigon and triangle moves to “clear” all the arcs that cross this 3-loop, which creates a 3-gon. This 3-gon corresponds to 3 squares in M around a vertex and L_1, L_2 and L_3 are leaves that cross the edges of these squares. By the completeness of the local cubulation, there exists a 3-cube of which these three squares are faces. The three leaves L_1, L_2, L_3 now meet in this 3-cube, a contradiction.

We now prove the claim for general n . We observe that a regular leaf itself inherits a web structure from M , but of smaller rank. Thus, we induct on the $rank(M)$ starting at $n = 3$. For $rank(M) = 1$, the leaves are points and the Proposition is clear. We assume that the claim is true for L_n . Note that for each $i < n$, $L'_i = L_i \cap L_n$ is a regular leaf in L_n . Moreover, for each $i, j < n$, by the statement of the claim for $n = 3$, we have that $L'_i \cap L'_j \neq \emptyset$. Thus, by induction, $\cap_{i=1}^{n-1} L'_i \neq \emptyset$. Since this intersection is a subset of L_n , we obtain that $\cap_{i=1}^n L_i \neq \emptyset$. $\square$

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As mentioned, the web structure on M induces a local cubulation structure on each regular leaf L , where the local cubes are the intersection of local cubes of M with L . In fact, we have the following.

Proposition 3.7. *Let M have a web structure and let L be a regular leaf of M . Then the induced local cubulation of L is a web structure.*

Proof. We need to check the following the following properties:

1. The local cubulation is CAT(0).
2. The local cubulation of L satisfies the 3-cube condition.
3. Limits of leaves in L are unique.

1. To see that the local cubulation is $CAT(0)$, consider $x \in L$. In M , x has a $CAT(0)$ cubulated neighborhood of the form $N = N' \times I$, where N' is a $CAT(0)$ cube complex. The regular cubulated neighborhood of x in L is thus N' , which is $CAT(0)$. This follows from the fact that the join of two flag simplicial complexes is flag if and only if each of the factors is flag.

2. Suppose that $\sigma_1, \sigma_2, \sigma_3$ is a cycle of three squares in L around a central point p . By the 3-cube property for M , there exists a 3-cube σ containing the σ_i 's as faces. Note that if σ_1 were local cubes then by the $CAT(0)$ property for the local cubulation of L , we would have that $\sigma \subset L$. More generally, we note that σ is built up of local cubes which are products of local edges along the intersections of the σ_i 's. We proceed to show that each of these local cubes are contained in L , starting with the local cube at $\cap_i \sigma_i$ and working out from there.

Maybe
more
-ms
put in
details...

3. Suppose that N_i is a sequence of disjoint regular leaves in L which converges to a leaf N in L . We wish to show N is connected. Note that each N_i is the intersection $L_i \cap L$ of a regular leaf L_i in M with L . Since $N_i \rightarrow N$, the sequence of leaves (L_i) converges to some leaf L' and by assumption, L' is connected. Now $N = L \cap L'$ and by Lemma 3.6, we have that N is connected. $\square$

4 Limit leaves

Having established some useful properties for regular leaves, we now discuss limits of regular leaves, which we will need in the construction of the median structure.

We consider a sequence (L_n) of regular leaves, whose intersection with some cube σ is a nested sequence of parallel plaques π_n which converge to a plaque π . (By nested, we mean that in σ , the plaque π_n separates π_{n-1} from π_{n+1} .) The limiting plaque π may be a boundary cube of σ . We will extend π to a limiting object to which L_n will converge on compact sets.

To this end, let $K \subset M$ be a finite union of maximal cubes. Note that for each n , the intersection $L_n \cap K$ is a regular leaf and hence contains no singular points of M . Since there are finitely many singular point in K , we may choose N sufficiently large, such that for all $n, m > N$, there exist no singular points of K between L_n and L_m . This will mean that for all $n > N$, the leaves $L_n^K = L_n \cap K$ cross the same edges and intersect each maximal cube of K in parallel plaques (or the intersection of σ with all such L_n 's is

empty), converging to a limiting plaque. The K -supported limit leaf L^K is the union of these limit plaques over all the maximal cubes of K . Note that this means that for each K , there exists a “half-regular neighborhood” for L^K . That is, there exists a product neighborhood $N_K \cong L^K \times [0, 1] \subset K$, such that L^K is identified with $L^K \times \{0\}$ in this product.

Now we exhaust M by compact sets $M = \cup_i K_i$, where each K_i is of the above type. Note that the definition of the limiting plaque of a cube in K_i does not change in any larger K_j , so that the limiting plaques are well-defined. We let L be the union of these limiting plaques, and we call such a leaf a *limit leaf*. Recall that the definition of a web structure is that such a limiting leaf is unique. Or said differently, that the leaf L defined above is connected in that there exists a leaf path between any two points in L .

Note that the limiting L is determined by a single edge

which intersects L in a single point and which intersects some transversely every tail of the sequence of regular leaves (L_n) . More precisely, suppose that a limiting L is given and a is a point on L . As noted above, there is an edge (1-cube) $e \cong [0, 1]$, such that $a = \{0\}$ and for every cube σ containing e , the intersection $L \cap \sigma$ is a plaque π of σ *orthogonal* to e : $\sigma \cong \pi \times e$. For all but countably many points $x \in e$, we have a unique regular leaf transverse to e at that point. For any decreasing (in the identification of e with $[0, 1]$) sequence of such points $(x_n) \rightarrow 0$, we obtain a sequence of regular leaves (L_n) . Given any finite union of cubes K , there is some N , such that the L_n ’s are all parallel on K for all $n > N$. Since this is true for any two such sequences, we see that they must converge to the same limit L . So in fact, the edge e determines L .

Conversely, given an edge e in M and an endpoint a of e , we have a leaf transverse to e at a , namely the limit of any sequence of regular leaves transverse to e whose intersection with e converges to a . We will call this the *limit leaf orthogonal to e at a* .

One of the defining features of web structures is what we call *uniqueness of limits*. Namely, that given a sequence of regular leaves (L_n) converging to L , the sequence does not converge to any other leaf. An illustrative example of why this assumption is necessary, is the following complete local cubulation of the plane, in which a sequence of leaves converges to two leaves.

INSERT: Example of disconnected limit: 2-dimensional reeb-like picture

We now wish to show that limit leaves have properties shared by regular leaves. These require tweaking the arguments given in the regular case of the previous section.

Proposition 4.1. *Let M be a web structure. Let L be a limit leaf. Then*

1. *L intersects any cube of M in at most one plaque.*
2. *Each limit leaf inherits a local cubical structure from M*
3. *For $a \in L$ and N a cubulated neighborhood of a , $N \cap L$ is an isometrically embedded subcomplex of N (and hence is $CAT(0)$).*
4. *L is simply connected.*
5. *The intersection of any two limit leaves is connected.*
6. *The intersection of a limit leaf L of M and a regular leaf of M is a regular leaf of L .*
7. *The local cubical structure of a limit leaf L is a web structure on L .*

Remark. Note that a limit leaf does not separate M into two components, as is the case for regular leaves. For example, if $M \cong [0, 1]$ with its natural web structure (cubes are closed subintervals and leaves are points), then $\{0\}$ and $\{1\}$ are limit leaves and all other points are regular leaves.

Proof. In general, the proofs proceed as for regular leaves discussed in the previous section, so we just highlight the differences.

1. Let L be a limit leaf. Suppose that L intersects a cube in two plaques. As in the regular case there are two possibilities.

One is that there is a local cube, in which two plaques intersect. The difference from the regular case, is that the two plaques could be intersecting faces of a local cube. However, the argument proceeds exactly as in the regular case. One finds a local square σ and boundary edges e and f in L which intersect at a vertex v of σ . Then we assume there is a piecewise straight leaf loop based at v , starting with e and ending with f . Then one finds a disk diagram and reduces the length or the area of the disk diagram just as in the proof of Lemma 3.3.

The case that the two plaques do not intersect also proceeds as in the proof of Lemma 3.3 as well.

2. The cubes of local cubical structure for L are the intersection of L with the local cubes of M . As we saw these are codimension 1 plaques of cubes of M

3. The neighborhood N of a is a finite CAT(0) cube complex consisting of cubes all of which contain a . We claim that there exists a unique $a \in$

□

Proposition 4.2. *Let L be a limit leaf. Then L inherits a web structure by intersecting the cubes of M with L .*

Proof. Let L be a limit leaf. As for regular leaves, we must check the three properties listed in Proposition 3.7. As noted above, on each compact union of cubes in K , each limit leaf has a half regular neighborhood. As a consequence, L inherits a natural CAT(0) cubulation from M . Namely, the cubes σ of L are of the form $\sigma \times \{0\}$ where $\sigma \times [0, 1]$ is a cube in M . As for regular leaves, this is a CAT(0) cubulation.

We must see that L has a local cubulation, that it satisfies the no-missing 3-cube condition and that leaves limit to a connected leaf. Let (L_n) be a sequence of leaves converging to L .

Local cubulation. Consider a point $x \in L$. By construction, there exists an edge e with x as an endpoint, such that every L_n is transverse to e . Note that regular leaves are locally CAT(0) cubulated since it follows from the facts that hyperplanes are CAT(0) cubulated in CAT(0) cube complexes (see [?]). The local cubical structure of x at L is the same as the local cubical structure of L_n at x . Thus L is locally CAT(0) cubulated as required.

No missing 3-cubes. If a triple of edges in L which span a 3-cube σ , then σ is also contained in L . Thus a triple of squares with a missing 3-cube in L , would give such a configuration in M , a contradiction.

might want more details here -ms

Connected limit. First suppose that a sequence of leaves L_n in L converge to disconnected L' . the sequence L_n

□

Remark. Note that a limit leaf may not separate M . For example consider the upper Euclidean halfplane squared in the standard way and a sequence of leaves of leaves $L_n = \{(x, 1/n) | x \in R\}$. Then the limit leaf is the x -axis which does not separate.

5 Constructing the median structure

The construction here follows what was done in dimension 2. Let M be an ER homology manifold with a web structure. We define $\mathcal{L}$ to be the collection of leaves (regular and singular, which are limits of regular leaves) of the web structure. Each leaf has two complementary components. A *half space* is the closure of one of the complementary components. We let $\mathcal{H}$ denote the collection of half spaces. We let $\mathcal{U}$ denote the collection of DCC ultrafilters on $\mathbb{H}$. Technical point: we want ultrafilters defined on all but possibly finitely many halfspaces; when α is a principal ultrafilter defined by a point $x \in M$, we don't pick a side for leaves that contain x .

not sure if we want closure or not. details of proof will make it clear –ms

Properties of leaves that are needed:

1. Leaves are embedded
2. Any two leaves intersect in a leaf of the web structure of each leaf (each leaf inherits a local cubulation structure)
3. Three leaves that pairwise intersect have a nontrivial triple intersection (this ensures that the intersection pattern of leaves inside a leaf agrees with the intersection pattern of the leaves in the ambient space).

The proof follows what is done in dimension 2 (in top median diary).

Lemma 5.1. *The intersection of a nested collection of halfspaces is either empty or a halfspace.*

Proof. Import from diary and adjust for higher dimensions □

One constructs a map

$$M \rightarrow \mathcal{U}$$

$$x \mapsto \alpha_x$$

where $\alpha_x = \{\mathfrak{h} \in \mathcal{H} \mid x \in \mathfrak{h}\}$. We now claim that this map is a bijection.

Lemma 5.2. *For each DCC ultrafilter $\alpha \in \mathcal{U}$, there exists a unique $x \in M$ such that $\alpha = \alpha_x$.*

Proof. Start with a sequence of nested halfspaces in α , their intersection is again a halfspace, so we can apply zorn to obtain a halfspace $\mathfrak{h}_0 \in \alpha$ such that for all leaves $\ell \in \mathfrak{h}_0$, the halfspace chosen by α is the complementary halfspace of the halfspace contained in $\mathfrak{h}_0$ associated.

Now we are left to consider the remaining leaves for which we have not yet chosen a side. These are all the leaves that meet ℓ_0 , the leaf associated to $\mathfrak{h}_0$. By induction the web structure on ℓ_0 has a median structure, so that a DCC ultrafilter on ℓ_0 determines a point on ℓ_0 . Since leaves of the web structure of ℓ_0 are intersections with leaves in M , a DCC ultrafilter on ℓ_0 makes a decision for all the remaining leaves (except for finitely many). $\square$

6 Surfaces

Web structure $\iff$ quartic differentials